

Foundations of Prompt Engineering & API Access

Part 2



Session 2 Agenda



Quick Recap - Session 1 Key Learnings

5 Applied Prompt Patterns

Real-World Use Cases

EXTRACTION TASKS

Description: Extract specific information structured as data points.

-  Extract entities (people, locations, organizations)
-  Identify key facts and events
-  Capture dates and relationships

BEFORE: Unstructured Input Text

Project meeting with Acme Corp. and team leads, set for next Tuesday, November 12th at 2PM, focused on Q4 budget. Sarah Chen attending.

AFTER: Extracted Structured Output

Entity Type	Value
Project	Q4 Budget
Client	Acme Corp.
Date	Nov 12th
Time	2PM
Attendee	Sarah Chen

CLASSIFICATION TASKS

Description: Categorize data into predefined classes or labels.



Intent detection



Sentiment analysis



Routing decisions

Sentiment Analysis

BEFORE: Customer Feedback

The new app update is fantastic!
So much faster.

AFTER: Classified Sentiment

[POSITIVE]

Intent Detection

BEFORE: User Query

How do I reset my password?

AFTER: Classified Intent

[ACCOUNT_SUPPORT]

TRANSFORMATION TASKS

Description: Reformatting, distilling, or changing the style of input content while preserving core meaning.



Summarization

Translation

Reformatting

Style transfer

Summarization Example

BEFORE: Original Input Text

Yesterday, the city council held a lengthy meeting to discuss the new public transport initiative. Many citizens voiced strong support for expanded bus routes, though some raised concerns about funding. Ultimately, the council voted 7-2 in favor of the expansion plan, which is set to begin next year.

AFTER: Summarized Output

City Council approved a new public transport expansion plan starting next year, despite some funding concerns.

Translation Example

BEFORE: English Phrase

Good morning! Where is the nearest train station?

AFTER: Translated Phrase (Spanish)

¡Buenos días! ¿Dónde está la estación de tren más cercana?

GENERATION TASKS

Description: Generating new, coherent content based on input parameters.

- ✍ Writing reports
- ✍ Generating code
- ✍ Creating SQL queries
- ✍ Creative content

SQL Query Generation Example

BEFORE: User Prompt

Get all customer emails from New York.

AFTER: Generated Query

```
SELECT email FROM customers  
WHERE state = 'NY';
```

Creative Content Generation Example

BEFORE: Topic/Constraints

Write a short poem about wind turbines in a field.

AFTER: Generated Poem

Giants stand, their arms they turn,
Catching breezes, a silent lesson to learn,
Power flows from the spinning rotation,
Fuelling a future, for a green nation.

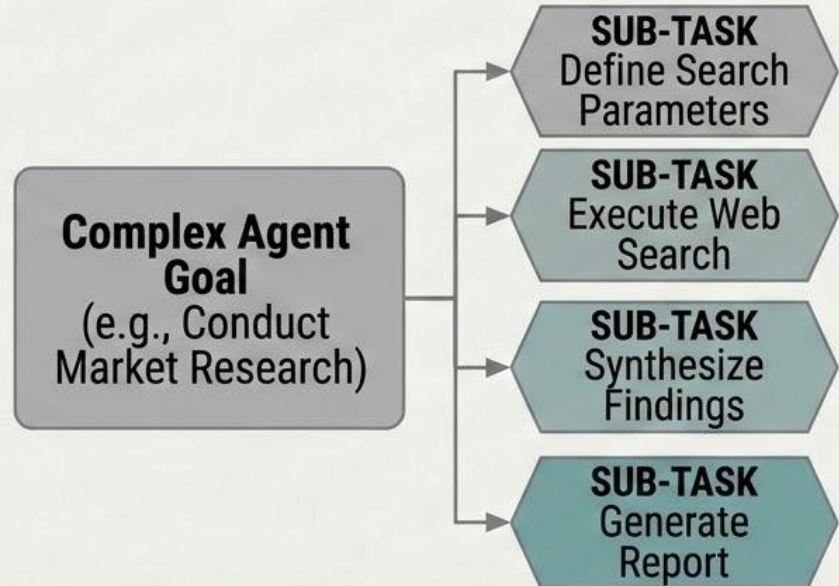
DECOMPOSITION

Description: Breaking down a complex task into smaller, manageable sub-tasks.

Why it powerful for agents:

- ☑ Tames Complexity
- ☑ Improves Explainability
- ☑ Facilitates Planning
- ☑ Enhances Error Handling

DECOMPOSITION FLOW



Advanced Reasoning Techniques

Moving Beyond Basic Prompting.

CHAIN OF THOUGHT PROMPTING

THINK STEP-BY-STEP

BASIC PROMPT (BEFORE)

Question: If there are 3 baskets, each with 5 apples, and I take 2 apples from one basket, how many apples are left in total?

Output: 13



COT PROMPT (AFTER)

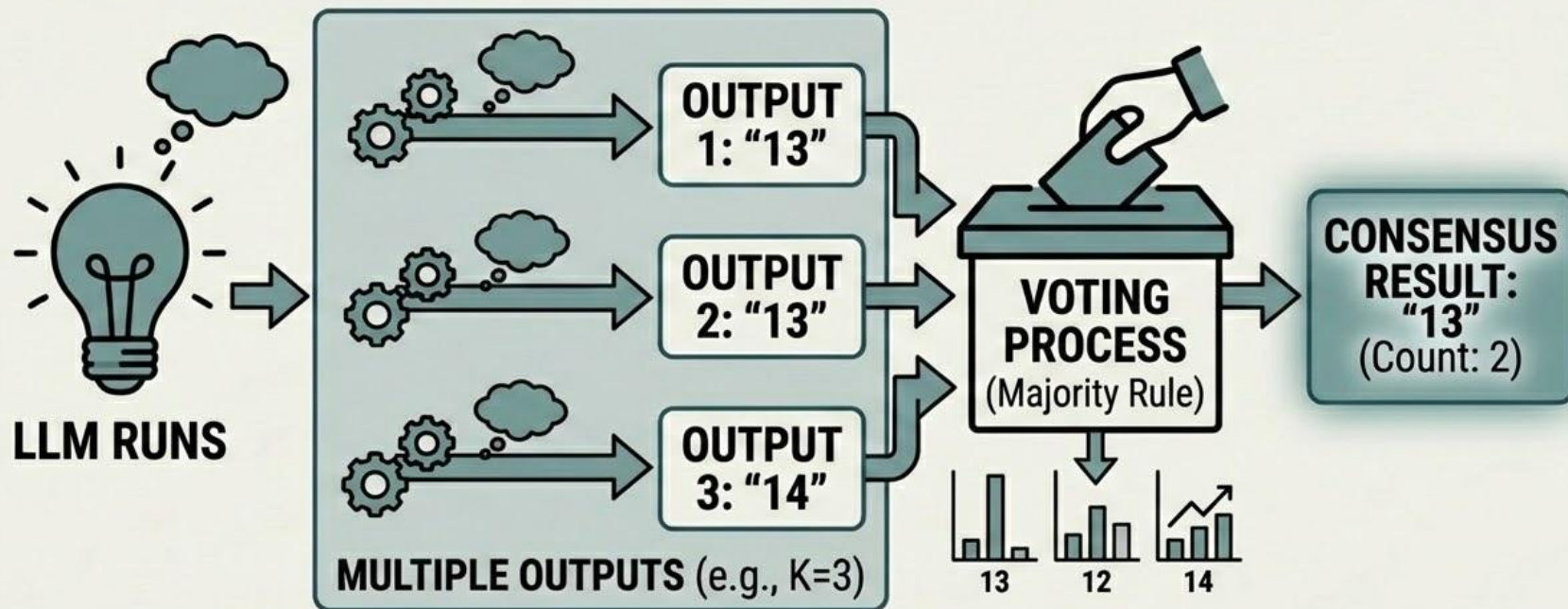
Instruction: If there are 3 baskets, each with 5 apples, and I take 2 apples from one basket, how many apples are left in total? Think step-by-step.

Improved Output: First, find the initial total number of apples: $3 \text{ baskets} * 5 \text{ apples/basket} = 15 \text{ apples}$.
Next, calculate the remaining apples:
 $15 \text{ initial apples} - 2 \text{ apples taken} = 13 \text{ apples}$. Total apples left: 13.

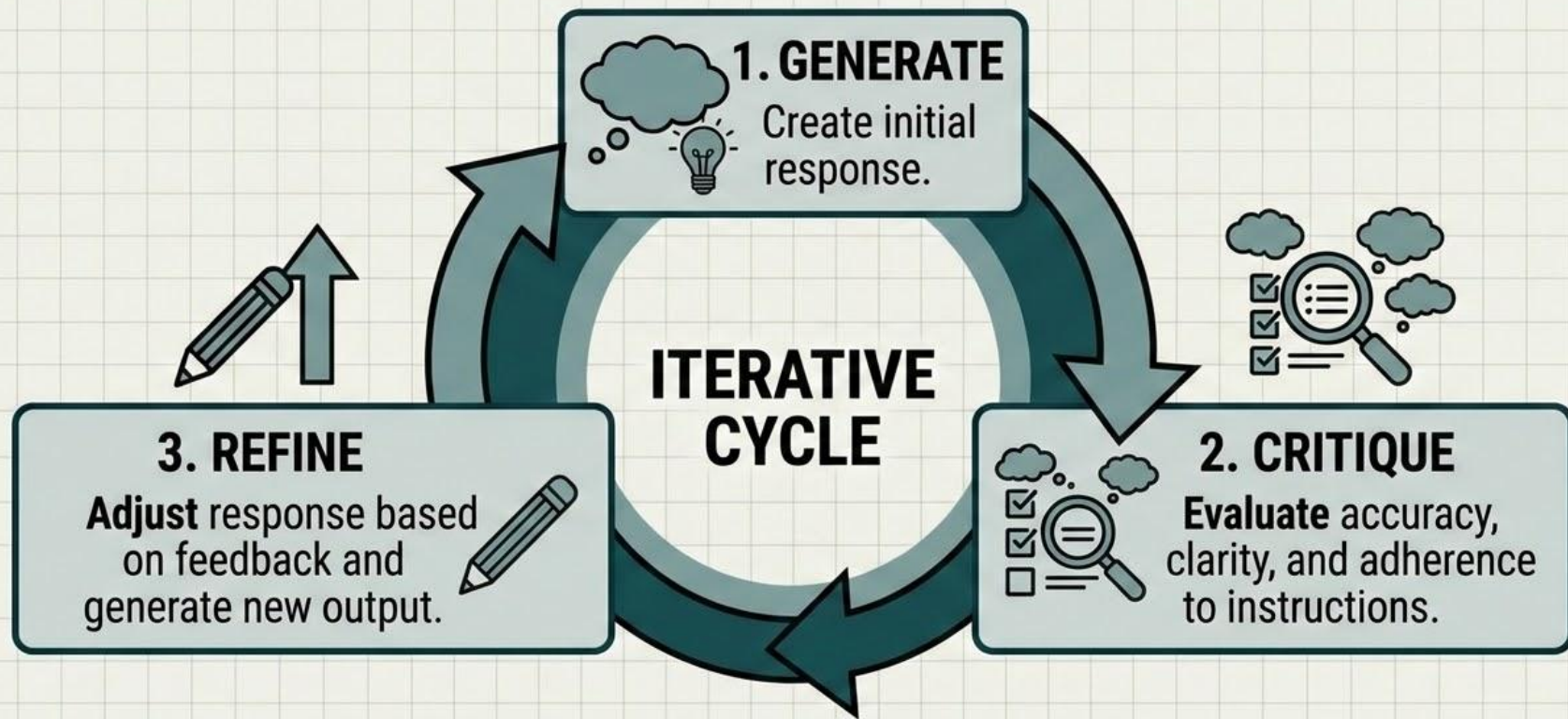


SELF-CONSISTENCY TECHNIQUE

Generate multiple responses and take majority vote.

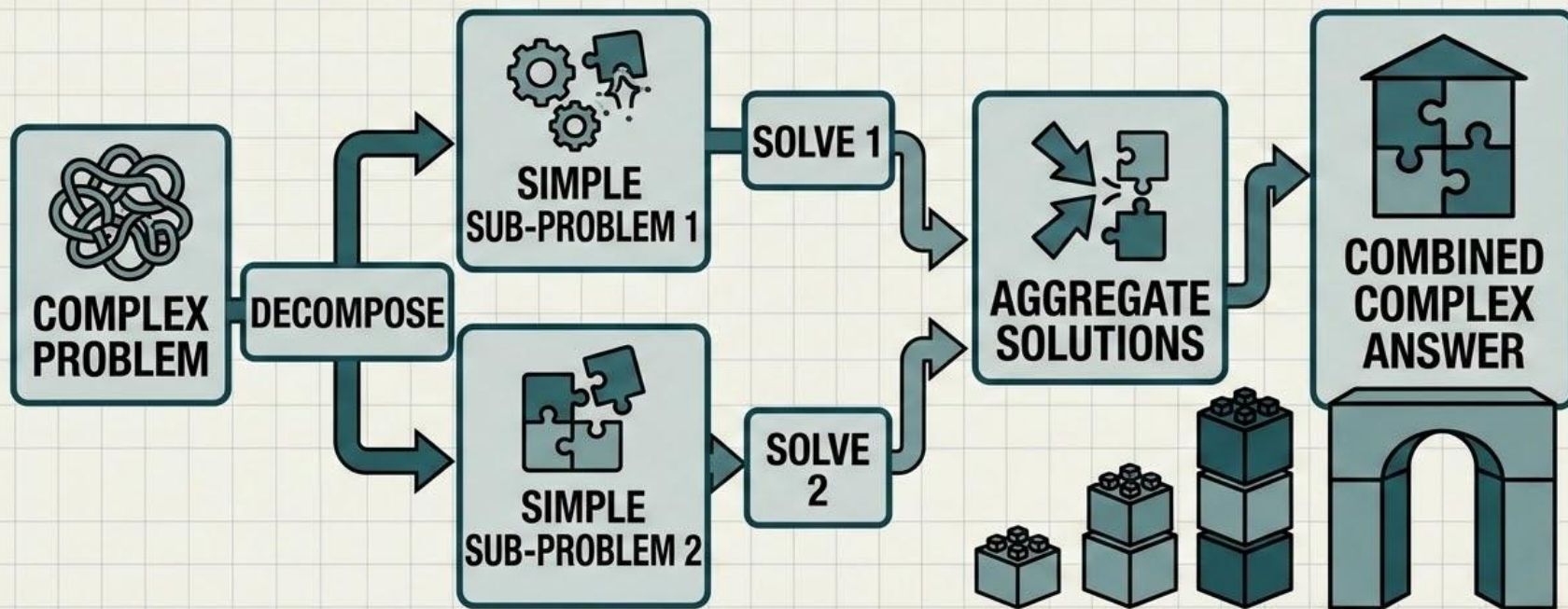


SELF-REFINE / CRITIQUE & IMPROVE



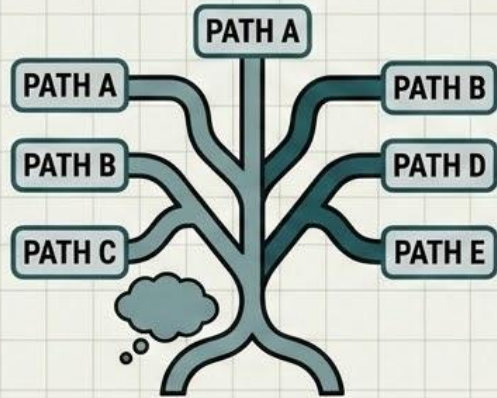
LEAST-TO-MOST PROMPTING

Solve simple sub-problems first, then combine. Good for complex reasoning.



TREE OF THOUGHTS (TOT)

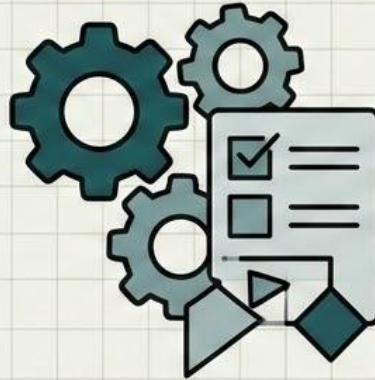
BRIEF OVERVIEW



Explore multiple reasoning paths like a tree.

**STRUCTURED
EXPLORATION.**

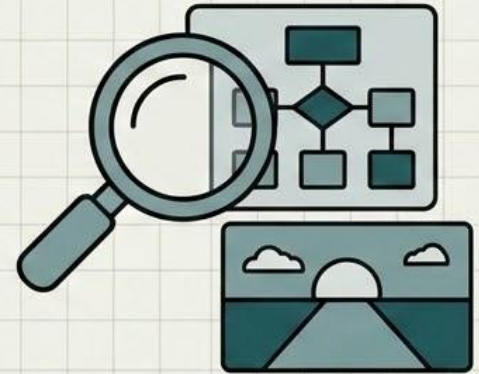
WHEN TO USE IT



Use when problem has many potential solutions.

**DECISION SUPPORT
& PLANNING.**

KEEP IT HIGH-LEVEL



Focus on the macro view and overall strategy.

**STRATEGIC
OVERSIGHT.**

7 PROMPT MANAGEMENT & PRODUCTION REALITIES



1. LATENCY & COST OPTIMIZATION

Balancing API speed vs. performance and budget constraints.



2. PROMPT VERSION CONTROL

Tracking prompt iterations, model variations, and avoiding regression.



3. GOVERNANCE & COMPLIANCE

Enforcing security, data privacy (e.g., GDPR), and ethical guidelines.



4. ERROR HANDLING & FALLBACKS

Designing robust retry logic, model fallback systems, and graceful failures.



5. AUTOMATED TESTING

Setting up scalable testing frameworks for prompt accuracy and model consistency.



6. HUMAN-IN-THE-LOOP (HITL).

Integrating expert review and continuous learning at scale.



7. SCALABILITY & CONCURRENCY.

Managing high throughput, variable load, and concurrent user requests.

PROMPT VERSIONING STRATEGIES



CODE-BASED VERSIONING

Version Control in Source Code

*PROS:

- Full granular control.
- Uses standard developer workflows (Git).
- Easy offline experimentation.

*CONS:

- Complex for non-technical stakeholders.
- Harder visual comparisons.
- Manual process to deploy to models.



MANAGED PLATFORM VERSIONING

Prompt Management SaaS

*PROS:

- Intuitive collaborative workspace.
- Side-by-side visual diffing.
- One-click deployment to various models.

*CONS:

- Vendor lock-in.
- Recurring subscription costs.
- Less integration with local development.

A/B TESTING PROMPT VARIANTS



1. DEFINE OBJECTIVE & METRICS

Set clear goals (e.g., Conversion, Clarity, or Accuracy) and determine success metrics.

- Define Target Metric
- Establish Baseline
- Success Criteria



2. DEVELOP PROMPT VARIANTS

Formulate hypotheses and create variations of the existing prompt (Prompt A) to test (Prompt B).

- Hypothesis Generation
- Control vs. Variable
- Systematic Changes



3. RUN THE EXPERIMENT

Randomly split production traffic and deliver different prompts to users to gather performance data.

- Data Collection
- Random Allocation
- Statistical Power



4. ANALYZE & ITERATE

Evaluate the results, determine the statistically significant winner, and the successful variant.

- Win Criteria (P-Value)
- Performance Comparison
- Rollout Winner

PROMPT CACHING AND COST CONTROL



CACHING BENEFITS

Reducing API expenses and improving efficiency

- **REDUCED LATENCY:** Faster response times.
- **SIGNIFICANT COST SAVINGS:** Lower token consumption.
- **DETERMINISTIC RESPONSES:** Consistent outputs.
- **IMPROVED SYSTEM PERFORMANCE:** Less strain on infrastructure.



COST OPTIMIZATION TECHNIQUES

Strategies for efficient prompt engineering

- **SEMANTIC CACHING:** Reuse prompts with similar intent.
- **TTL MANAGEMENT:** Set appropriate cache duration.
- **PROMPT STANDARDIZATION:** Maintain consistent prompt formats.
- **TOKEN MINIMIZATION:** Use concise instructions and inputs.

LIVE EXERCISE: BUILDING A MINI AGENT



1. DEFINE OBJECTIVE

Specify the complex task for the mini-agent (e.g., summarizing research on a new technology).

- Set clear primary goal
- Define success criteria
- Identify necessary data inputs



2. DECOMPOSITION

Break the complex goal into distinct, manageable sub-tasks.

- Identify key sub-problems
- Create logical task ordering
- Define output requirements per step



3. COT REASONING

Design a multi-step reasoning prompt using Chain-of-Thought logic for each sub-task.

- Use detailed instructions
- Incorporate 'let's think step by step'
- Structure inputs for transparency



4. EXECUTE & VALIDATE

Run the full multi-step prompt and analyze the performance.

- Evaluate final output clarity
- Validate reasoning correctness
- Plan iterations based on results

KEY TAKEAWAYS



- **MASTER AGENT DESIGN:** Structure complex tasks using decomposition and Chain-of-Thought reasoning.
- **IMPROVE EFFICIENCY & REDUCE COST:** Apply prompt caching strategies (semantic, TTL) and minimization techniques.
- **ITERATE AND VALIDATE:** Analyze multi-step prompt performance rigorously for consistency and correctness.



Thank You for Attending!
Questions?